

Wolfpack Agency — Complete Infrastructure & Cost Report

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Executive Summary

This report covers the complete infrastructure costs for both Wolfpack products:

- **Wolfpack Instinct** — Team intelligence platform (assistant, knowledge base, reports, analytics, briefings, financials, settings)
- **Wolfpack Auto** — Dealer management platform (inventory, leads, F&I, payments, payroll)

Total monthly cost for both platforms combined: ~\$120/month (range \$107–124).

METRIC	VALUE
Wolfpack Instinct (monthly)	\$100.50–102
Wolfpack Auto (monthly)	\$6–22
Combined annual estimate	\$1,284–1,488

Key change vs. April 6 report: the team Assistant now runs on GPT-4o-mini (hybrid model approach) instead of Claude. The CTO development tool (Claude Max, \$100/mo) is unchanged because it is the right tool for agentic coding workloads. Net savings: \$5–15/mo today, with much larger implications for Wolfpack Auto at scale (see "Why hybrid matters" below).

Why Hybrid LLM (Not Single-Vendor)

After evaluating single-vendor approaches, we adopted a **per-workload model** strategy:

WORKLOAD	MODEL	REASON
CTO development tool (Claude Max, IDE coding loop)	Claude Opus 4.6	Strongest agentic coding model. Used by one person. Fixed cost.

WORKLOAD	MODEL	REASON
Wolpack Instinct team Assistant (5 users, Q&A + briefings)	GPT-4o-mini	Cheapest credible model. Zero-token-first cache absorbs ~80% of queries before any LLM call.
Wolpack Instinct premium queries (rare, complex reasoning)	GPT-4o	Mid-tier reasoning when mini isn't enough. <5% of traffic.
Wolpack Auto embeddings	OpenAI text-embedding-3-small	Already in production. Pennies per month.
Wolpack Auto LLM features (future)	GPT-4o-mini for templated content, GPT-4o for customer-facing	Multi-tenant unit economics demand the cheapest model that meets quality bar.

The hybrid approach is not primarily a cost decision today. Today's savings are real but small (~\$5–15/mo). The real value is unit-economics protection for Wolpack Auto at scale and enterprise positioning for Instinct (Azure OpenAI option keeps client data inside the same M365 tenant). See the final section.

Platform 1 — Wolpack Instinct (Team Intelligence Platform)

AI-powered workspace for the entire Wolpack team (5 members): Assistant, knowledge base, reports, discussions, feature requests, journal, client management, email templates, analytics, morning briefings, financials (CEO-only), and Microsoft 365 + QuickBooks integrations.

Monthly Costs

SERVICE	PURPOSE	PLAN	MONTHLY	ANNUAL
Claude Max	Nick's development tool (IDE, Claude Code, building Instinct + Auto)	Max 5x (Personal)	\$100.00	\$1,200.00
OpenAI API (GPT-4o-mini)	Powers the Wolpack Assistant + morning briefings for all 5 team members	Pay-as-you-go	\$0.50 – \$2.00	\$6 – \$24
Vercel	Hosting (wolpack-instinct.vercel.app)	Hobby (Free)	\$0.00	\$0.00

SERVICE	PURPOSE	PLAN	MONTHLY	ANNUAL
Neon PostgreSQL	Primary database	Free	\$0.00	\$0.00
Qdrant Cloud	Vector search for knowledge matching	Free	\$0.00	\$0.00
Neo4j AuraDB	Knowledge relationship graph	Free	\$0.00	\$0.00
Resend	Email notifications (3,000/mo)	Free	\$0.00	\$0.00
Sentry	Error monitoring	Free	\$0.00	\$0.00
Microsoft Graph API	Calendar / email / contacts ingest (per-user OAuth)	Free	\$0.00	\$0.00
QuickBooks Online API	Financial data (CEO-only)	Free	\$0.00	\$0.00
Instinct Subtotal			\$100.50 – \$102	\$1,206 – \$1,224

Why the team Assistant is so cheap

Three reinforcing reasons the LLM line is sub-\$2/month for a 5-person team:

1. **Zero-token-first architecture.** Every Assistant query first checks the cached knowledge base, then Qdrant semantic search, then Neo4j graph context. Roughly 80% of questions are answered without ever calling an LLM. The LLM is the fallback, not the default.
2. **GPT-4o-mini pricing.** \$0.15 input / \$0.60 output per million tokens. Five users at 20 messages/day with ~70% cache hit rate uses approximately 3 million tokens/month total — about \$0.50.
3. **Per-LLM-call cost cap.** Every Assistant response logs token usage to analytics. If usage spikes, we have per-user budget enforcement ready to deploy from the existing analytics events.

What every team member gets

- Wolfpack Assistant — AI-powered Q&A that learns from every interaction
- Knowledge Base — searchable team knowledge, compounds over time
- Morning Briefing — calendar + email + action items pulled from M365 (per-user, isolated)
- Reports — branded client reports, proposals, internal audits
- Discussions — threaded team conversations with resolution tracking
- Feature Requests — submit ideas, vote, track development
- Journal — team activity log + daily context

- Client Management — client profiles + communication tracking
- Email Templates — professional client emails generated instantly
- Analytics — usage insights, team activity, AI efficiency metrics
- Settings / Integrations — connect personal Microsoft 365 + (CEO) QuickBooks
- Financials (*CEO only*) — P&L, balance sheet, AR/AP aging, invoices

Platform 2 — Wolfpack Auto (Dealer Management Platform)

Full dealer operations platform: inventory management, lead tracking, F&I, payments, payroll, compliance, analytics, and customer-facing dealer websites.

Fixed Monthly Costs (Always On)

SERVICE	PURPOSE	PLAN	MONTHLY	ANNUAL
Vercel	Hosting dealer platform	Hobby (Free)	\$0.00	\$0.00
Neon PostgreSQL	Dealer data, leads, vehicles, deals, payroll	Free	\$0.00	\$0.00
Cloudflare R2	Vehicle images and media storage	Free (10 GB/mo)	\$0.00	\$0.00
Qdrant Cloud	Semantic vehicle search	Free	\$0.00	\$0.00
Neo4j AuraDB	Customer journey + relationship tracking	Free	\$0.00	\$0.00
Resend	Lead notifications, invoices (3,000/mo)	Free	\$0.00	\$0.00
Sentry	Error monitoring	Free	\$0.00	\$0.00
Google Maps	Dealer location maps	Free (\$200 credit)	\$0.00	\$0.00
NHTSA API	VIN decoding	Free (federal)	\$0.00	\$0.00
Let's Encrypt	SSL certificates	Free	\$0.00	\$0.00
NextAuth.js	Authentication	Free (open source)	\$0.00	\$0.00

SERVICE	PURPOSE	PLAN	MONTHLY	ANNUAL
OFAC / Treasury	Sanctions compliance screening	Free (federal)	\$0.00	\$0.00
Fixed Subtotal			\$0.00	\$0.00

Variable Costs (Usage-Based)

SERVICE	PURPOSE	PRICING	EST. MONTHLY	EST. ANNUAL
Stripe	Payment processing	2.9% + \$0.30/txn	Varies*	Varies*
Twilio	SMS lead outreach	\$0.0079/SMS	\$5 – \$15	\$60 – \$180
FAL.ai	AI background removal on vehicle photos	Pay-per-use	\$1 – \$5	\$12 – \$60
OpenAI	Text embeddings for inventory search	\$0.02/1M tokens	\$0 – \$2	\$0 – \$24
Variable Subtotal			\$6 – \$22	\$72 – \$264

* Stripe fees only occur on actual transactions and are typically passed through to dealer or customer. There is no monthly fee.

Combined Monthly & Annual Totals

CATEGORY	MONTHLY	ANNUAL
Wolfpack Instinct (Team Platform)		
Claude Max (Nick's development)	\$100.00	\$1,200.00
OpenAI API — GPT-4o-mini (team assistant)	\$0.50 – \$2.00	\$6 – \$24
Databases, hosting, email, monitoring	\$0.00	\$0.00
Wolfpack Auto (Dealer Platform)		

CATEGORY	MONTHLY	ANNUAL
All fixed infrastructure	\$0.00	\$0.00
Twilio SMS (estimated)	\$5 – \$15	\$60 – \$180
FAL.ai image processing (estimated)	\$1 – \$5	\$12 – \$60
OpenAI embeddings (estimated)	\$0 – \$2	\$0 – \$24
Stripe (transaction fees, pass-through)	Varies	Varies
Grand Total (excl. Stripe pass-through)	\$107 – \$124	\$1,284 – \$1,488

Realistic monthly cost: ~\$120/month.

19 of 23 services run on free tiers. The remaining lines are either fixed (Claude Max, the CTO development tool) or transaction-based (Twilio SMS, FAL.ai, OpenAI embeddings, Stripe) and only incur charges when the platforms are actively generating value.

What Changed vs. the April 6 Report

LINE	APRIL 6 (CLAUDE-ONLY)	APRIL 7 (HYBRID)	DELTA
Wolfpack Instinct — team Assistant LLM	\$5 – \$15 (Claude API)	\$0.50 – \$2 (GPT-4o-mini)	–\$4.50 to –\$13
Wolfpack Instinct — Claude Max (CTO)	\$100	\$100	unchanged
Wolfpack Auto — OpenAI embeddings	\$0 – \$2	\$0 – \$2	unchanged
Everything else	unchanged	unchanged	unchanged
Total	\$111 – \$137	\$107 – \$124	–\$5 to –\$15/mo

Annual savings: ~\$60 to ~\$180. Today.

Why Hybrid Matters (Beyond Today's Savings)

Today's savings are small. The architectural decision matters for two things this report cannot fully capture yet:

1. Wolfpack Auto unit economics at scale

The current \$0–2/mo OpenAI line on Wolfpack Auto is **embeddings only**. There are no LLM-powered dealer features in production yet. When those ship (VDP descriptions, customer chat, lead intelligence narratives), the per-dealer cost becomes the binding constraint on the business.

Projected monthly LLM cost at 50 dealers, by vendor strategy:

STRATEGY	APPROX. MONTHLY COST	PER-DEALER
Claude Opus everywhere	\$1,800 – \$2,500	\$36 – \$50
Claude Sonnet everywhere	\$400 – \$550	\$8 – \$11
GPT-4o everywhere	\$300 – \$400	\$6 – \$8
Hybrid: GPT-4o-mini bulk + GPT-4o premium	\$60 – \$120	\$1.20 – \$2.40

This is the actual reason the architecture decision matters. At one internal dealer it's noise. At 50 paying dealers it is the difference between healthy margins and underwater unit economics.

2. Wolfpack Instinct enterprise positioning

For future client deployments of Instinct, the LLM provider can be configured as **Azure OpenAI** instead of OpenAI direct. Same models, same prices. The strategic difference: Azure OpenAI lives inside the customer's Azure / Microsoft 365 tenant — same Entra ID, same compliance boundary, no second vendor on a security review. For clients already on M365 (the majority of enterprise prospects), this collapses the procurement story from "we use a third-party AI vendor" to "your data never leaves your Azure tenant."

This is a sales-cycle asset, not a cost savings.

What's NOT Needed

The following are explicitly **not required** and have zero cost:

- M365 Copilot per-seat licensing (~\$30/user/mo) — Instinct provides the team-AI layer ourselves
- Individual Claude or ChatGPT subscriptions for team members — they use the Wolfpack Assistant through Instinct

- Separate chat tools or Slack subscriptions
- Additional Vercel seats (viewers are free)
- Redis / caching layer (built-in caching handles current scale)
- Custom domain SSL certificates (automatic through Vercel)

When Costs Would Increase

TRIGGER	WHAT CHANGES	ADDED COST
Database exceeds 0.5 GB (either platform)	Upgrade Neon to Launch	+\$19/mo
Over 3,000 emails/month	Upgrade Resend to Pro	+\$20/mo
Custom email domain (@thewolfpack.agency)	Resend Pro	+\$20/mo
1,000+ vehicles in inventory	Elasticsearch for advanced search	+\$16–95/mo
High site traffic (1,000+ daily)	Vercel Pro + Redis	+\$20–30/mo
Managed payroll for dealers	Gusto integration	+\$40+/mo
Wolfpack Auto launches LLM dealer features	OpenAI mini + 4o (per-dealer scaling)	~\$1–2 per dealer/mo

None of these (except eventual dealer LLM features) are expected at launch or in the near term.

Steps to Go Live

Wolfpack Instinct

1. Set environment variables in Vercel (API keys, database URLs, **OpenAI API key**, Microsoft Graph credentials, QuickBooks credentials)
2. Run database migrations (one command — includes per-user MS Graph isolation migration shipped April 7)
3. Create production accounts for each team member
4. Each team member connects their own Microsoft 365 account via Settings → Integrations
5. CEO connects QuickBooks Online via Settings → Integrations

Wolfpack Auto

1. Set environment variables in Vercel (database, Stripe, Resend, Twilio)
2. Run database migrations
3. Create production admin user and first dealer account
4. Import initial vehicle inventory

Estimated time to production: under 2 hours for both platforms.

Wolfpack Agency | Confidential | April 7, 2026 | Revision: hybrid LLM model